

BO I VESTERALEN, Norway

WMO: 011560

Lat: 68.607N

Lon: 14.433E

Elev: 8

StdP: 101.23

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.5	-5.3	-12.6	1.3	-5.4	-11.2	1.4	-4.5	20.2	3.2	18.0	1.8	4.5	70	0.787

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.8	20.6	15.0	18.7	14.0	17.0	13.2	15.7	19.3	14.6	17.8	13.6	16.4	3.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	10.0	17.1	13.0	9.3	15.6	12.2	8.9	14.7	43.8	19.3	40.8	17.9	38.4	16.5	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	14.1	12.3	-8.2	23.9	1.5	2.3	-9.3	25.6	-10.1	26.9	-11.0	28.3	-12.0	30.0		
(5)			-9.4	17.5	1.3	1.6	-10.3	18.6	-11.1	19.6	-11.9	20.5	-12.8	21.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	0.5	0.0	0.5	3.2	6.9	9.4	12.7	12.4	10.0	6.3	3.7	2.1		
	DBStd	5.24	2.99	2.94	2.83	2.58	2.92	2.79	2.84	2.50	2.52	2.86	2.82	2.77		
	HDD10.0	1808	294	279	294	205	104	43	3	5	29	118	188	244		
	HDD18.3	4622	553	512	553	454	354	269	179	185	249	373	438	502		
	CDD10.0	233	0	0	0	1	8	25	86	79	31	3	0	0		
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0		
	CDH23.3	8	0	0	0	0	0	1	7	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH27.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	5.6	6.5	6.5	6.6	5.4	5.0	4.7	4.0	4.1	5.0	6.0	6.6	7.0		
(15) Precipitation	PrecAvg	1104	97	94	90	69	63	54	62	75	108	144	123	134		
	PrecMax	1372	227	208	170	119	110	128	170	202	232	261	310	239		
	PrecMin	839	0	48	15	16	19	0	8	29	37	28	0	9		
	PrecStd	160	54	36	42	25	24	32	41	37	47	62	70	62		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	6.3	7.1	13.6	18.5	21.6	24.1	21.3	17.6	13.2	10.4	8.1		
		MCWB	5.5	5.0	4.5	8.3	12.9	14.3	18.1	15.7	12.9	9.8	9.0	6.8		
	2%	DB	6.2	5.6	6.0	10.0	15.0	18.3	21.7	19.4	16.0	12.0	9.2	7.4		
		MCWB	4.8	4.5	4.5	6.2	10.2	13.0	15.9	14.7	12.1	9.3	7.8	6.2		
	5%	DB	5.6	5.0	5.2	7.5	13.0	15.7	19.5	18.0	14.7	10.9	8.5	6.8		
		MCWB	4.2	3.9	3.9	4.9	9.2	11.4	14.6	14.1	11.4	9.2	6.9	5.5		
	10%	DB	4.7	4.3	4.6	6.4	11.1	13.7	17.3	16.4	13.5	10.0	7.7	6.1		
		MCWB	3.3	3.0	3.2	4.4	7.4	10.5	13.7	13.3	11.0	8.5	6.1	4.4		
	0.4%	WB	6.1	5.7	5.5	8.6	13.0	15.0	19.1	16.7	13.7	10.9	9.5	7.5		
		MCDB	6.6	6.0	6.0	13.5	18.3	20.2	23.6	19.5	16.6	12.1	10.2	7.8		
	2%	WB	5.2	4.8	4.7	6.5	10.8	13.4	16.6	15.3	12.6	10.2	8.3	6.7		
		MCDB	6.1	5.5	5.6	8.9	14.4	17.6	20.6	18.4	14.8	11.3	9.1	7.3		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	4.4	4.1	4.2	5.6	9.2	12.0	14.9	14.5	12.1	9.4	7.3	5.7		
		MCDB	5.5	5.0	5.3	7.2	12.4	15.1	18.6	17.3	14.0	10.6	8.3	6.7		
	10%	WB	3.4	3.2	3.3	4.9	8.1	10.9	13.9	13.6	11.5	8.7	6.3	4.7		
		MCDB	4.6	4.2	4.5	6.2	10.5	13.4	16.8	15.9	13.1	10.0	7.6	5.9		
	5% DB	MDBR	3.1	3.2	3.4	3.9	4.5	4.3	4.8	4.4	3.8	3.1	3.2	3.2		
		MCDBR	3.3	3.5	3.1	5.2	7.4	7.7	8.3	6.9	5.2	3.1	3.4	3.7		
(36) Mean Daily Temperature Range	5% WB	MCWBR	3.2	3.3	2.8	3.3	4.1	4.4	4.4	3.7	3.1	2.5	3.4	3.6		
		MCDBR	3.3	3.6	3.2	4.4	6.8	6.9	7.7	6.0	4.0	2.8	3.4	3.8		
	5% WB	MCWBR	3.4	3.7	3.0	3.3	3.9	4.3	4.5	3.5	2.9	2.7	3.7	3.9		
		MCDBR	3.4	3.7	3.0	3.3	3.9	4.3	4.5	3.5	2.9	2.7	3.7	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.371	0.293	0.327	0.342	0.346	0.344	0.348	0.341	0.327	0.295	0.352	N/A		
	taud		2.511	2.140	2.256	2.380	2.453	2.502	2.518	2.526	2.513	2.244	2.499	N/A		
	Ebn at Noon		74	466	691	804	845	858	841	806	715	462	64	0		
	Edh at Noon		11	45	74	87	91	89	85	76	59	39	9	0		
(44) All-Sky Solar Radiation	RadAvg		0.02	0.30	1.38	3.08	4.45	4.89	4.45	3.22	1.70	0.56	0.07	0.00		
	RadStd		0.00	0.07	0.20	0.27	0.34	0.70	0.46	0.30	0.16	0.10	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page